

FAISAL ASLAM

Senior AI/ML Engineer | RAG and Agentic AI Expert | AI Solution Architect

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SUMMARY

Senior AI/ML Engineer and AI Solution Architect with **6+ years** designing and shipping production Generative AI systems across Large Language Model (LLM) applications, agentic AI, and Computer Vision. Specialized in multi-agent orchestration (LangGraph, CrewAI), Retrieval-Augmented Generation (RAG, GraphRAG), tool-calling agents, and autonomous workflows deployed on AWS, GCP, and Azure. Currently Senior AI Engineer (Lead) at Tekhqs, owning end-to-end architecture, cloud infrastructure, evaluation, and cost optimization for client-facing AI platforms. Proven record of converting complex requirements into reliable, compliant (HIPAA, UK GDPR) AI products with measurable business impact.

TECHNICAL SKILLS

Languages: Python (async / asyncio), SQL, JavaScript, Rust (integration), Bash

AI and ML Frameworks: PyTorch, Hugging Face Transformers, PEFT, LangGraph, LangChain, CrewAI, LlamaIndex, n8n, Machine Learning, Deep Learning, Neural Networks, Model Training and Evaluation

Generative AI and LLMs: Large Language Models (LLMs), Anthropic Claude API, OpenAI API, Google Gemini API, Azure OpenAI, AWS Bedrock, GPT-4o, LLaMA 3.2, Qwen, GLM, Pixtral, Prompt Engineering, System Prompt Design, Prompt Caching, Context Engineering, Multimodal AI, Multi-LLM Orchestration, Model Routing and Fallback, Structured Outputs, Streaming Responses

Agentic AI and Orchestration: AI Agents, Agent Engineering, Multi-Agent Systems and Orchestration, Supervisor Graphs, Sub-Agent Decomposition, State Management, Checkpointing, ReAct, Planning Loops, Tool Calling, Function Calling, Tool Schema Design, Model Context Protocol (MCP), Browser and Computer-Use Agents, Agent Memory, Self-Correcting Agents, Human-in-the-Loop, AgentOps

RAG and Retrieval: Retrieval-Augmented Generation (RAG), Agentic RAG, Hybrid RAG, Hybrid Retrieval, GraphRAG, LightRAG, Embedding Models, Semantic Search, Chunking Strategies, Document AI, Table Extraction (Docling), Azure AI Document Intelligence, Natural Language to SQL (NL-to-SQL)

Databases and Vector Stores: PostgreSQL, MongoDB, Redis, Neo4j (Graph Database), pgvector, FAISS, Qdrant, Pinecone, Vector Databases, Amazon RDS, Amazon ElastiCache, Amazon S3, Database Design, Data Pipelines

Evaluation, Observability and LLMops: LangSmith, LLM Evaluation, Evaluation Frameworks (Evals), AI Observability, Structured Logging, Tracing, Benchmarking, Token Economics, Latency and Cost Monitoring, Prompt Regression Testing, LLMops, RAGops, GenAIops

Model Serving, Inference and Tuning: Self-Hosted LLM Serving, vLLM, FP8 Quantization, Mixture-of-Experts (MoE), GPU VRAM Sizing, OpenAI-Compatible Endpoints, Fine-Tuning (LoRA, QLoRA, PEFT), Model Benchmarking

Cloud, DevOps and MLOps: AWS (SageMaker, Bedrock, Lambda, EC2, ECS / Fargate, S3, IAM, VPC, API Gateway), GCP (Vertex AI, Cloud Run, Pub/Sub), Azure (Azure OpenAI, Azure AI Foundry), Docker, Terraform (Infrastructure as Code), MLflow, CI/CD, Git and GitHub, Linux, nginx, systemd, Serverless Deployment, MLOps

Backend, APIs and Full-Stack: FastAPI, Flask, Django, REST APIs, WebSockets, Server-Sent Events (SSE), Pydantic, Salesforce API, Google Maps API, Playwright, Backend and Frontend Development, Full-Stack Delivery, API and Database Design, End-to-End Product Ownership

Computer Vision, Voice and Speech AI: YOLOv8, DeepSORT, OpenCV, Object Detection and Tracking, Face Recognition, Image Segmentation, License Plate OCR, RTSP Video Streams, OpenAI Whisper (ASR), Speaker Diarization, Google STT / TTS, Real-Time Transcription, SIP and VoIP Integration

Web3 and Blockchain: Conversational Trading Agents, Solana, Base, World Chain, Sui, DeFi, Non-Custodial Wallet Integration, Client-Side Transaction Signing, On-Chain Tool Execution, DexScreener and Hyperliquid APIs

Responsible AI, Compliance and Governance: Guardrails, Deterministic Safety Checks, Human Oversight, Fallback Behavior, Graceful Degradation, Audit Trails, Data Residency, PII and PHI Handling, HIPAA, UK GDPR (Data Minimization, Consent Management, Right-to-Erasure), Role-Based Access Control (RBAC)

Leadership and Role Focus: Technical Team Leadership, Solution Architecture, Code Review, Mentoring, Agile Delivery, CTO and Executive-Level Technical Documentation, Cross-Functional Stakeholder Management. Role focus: Generative AI Engineer, Agentic AI Engineer, LLM Engineer, Machine Learning Engineer, Applied AI Engineer, AI Software Engineer, MLOps Engineer

WORK EXPERIENCE

Senior AI Engineer (Lead)

May 2026 - Present

Tekhqs | Lahore, Pakistan | Full-time

- Own end-to-end AI orchestration architecture for client-facing Generative AI platforms: multi-agent graph design, RAG pipelines, prompt and model optimization, deployment, and observability.
- Author CTO-level technical proposals (self-hosted LLM migration, cost and latency benchmarking, feature roadmaps) driving architecture and vendor decisions.
- Supervise 3 junior engineers, running code review, technical direction, and delivery oversight across concurrent AI workstreams.

Project: Kresus | Web3 Conversational AI Trading Agent (Multi-Chain DeFi Copilot)

Stack: LangGraph, Claude, FastAPI, Server-Sent Events, Solana, Base, World Chain, Sui, Tool Calling, Non-Custodial | Live: kresus.com

- Architected the AI orchestration layer for a production Web3 conversational trading agent: a FastAPI and LangGraph service driving Claude that lets users research, analyze, and execute on-chain trades across Solana, Base, World Chain, and Sui in natural language, replacing an in-process Rust agent.
- Engineered a multi-turn agentic tool-calling loop over 30+ trading tools (token discovery, DexScreener and Hyperliquid market data, technical analysis, swaps, transfers, portfolio balances, token deployment), streaming to the mobile app via Server-Sent Events (SSE).
- Preserved a non-custodial, client-side signing model in which tools return prepared, unsigned transactions for the mobile wallet to sign, and enforced deterministic unsupported-chain and cross-chain-swap guardrails in code.
- Reduced per-conversation LLM spend and time-to-first-token via prompt caching and system-prompt restructuring, and instrumented structured logging of input, intent, and tool-call signals for request-level observability.
- Specified a self-hosted open-weight LLM migration path (Qwen-class FP8 behind an OpenAI-compatible endpoint) to remove third-party API dependency, including the benchmark plan and effort estimate presented to the CTO.

Project: Vyera AI | Competitive Intelligence Multi-Agent Platform

Stack: LangGraph, Multi-Agent, Firecrawl, FastAPI, AWS, UK GDPR | Live: vyera.ai

- Architected an AI competitive intelligence platform for the UK market that autonomously discovers competitors, scrapes data via Firecrawl, and synthesizes strategic insights through a multi-agent LangGraph supervisor graph.
- Built specialized agents for web scraping, social media intelligence, pricing analysis, and market positioning, producing automated 360-degree competitor profiles that replace roughly 8 analyst hours of manual research per report.
- Engineered full UK GDPR compliance (data minimization, consent management, lawful-basis documentation, right-to-erasure) and owned end-to-end AWS infrastructure including IAM, VPC and security-group hardening.

Senior AI Engineer, Team Lead

Mar 2025 - May 2026

Linvox Solutions | Lahore, Pakistan | Full-time, On-site

- Led a cross-functional team of 5 engineers, owning solution architecture, code review, AWS infrastructure, and delivery across enterprise Generative AI projects.

Project: Advanced Hybrid RAG | Enterprise Knowledge Assistant (Saudi Aramco)

Stack: LightRAG, Neo4j Knowledge Graph, Azure AI Document Intelligence, LangGraph, GraphRAG | Live: auxee.com

- Built a hybrid multi-agent Retrieval-Augmented Generation system pairing LightRAG over a Neo4j knowledge graph for entity and relationship extraction with Azure AI Document Intelligence for structured parsing across all document formats.
- Implemented smart retrieval routing (LightRAG plus SQL for Excel/CSV; Azure Document Intelligence plus GraphRAG for PDF/DOCX/TXT), enabling multi-hop reasoning beyond vector-only search.
- Deployed internally at Saudi Aramco for petroleum-management teams, letting staff query large volumes of technical and operational documents in natural language.

Project: NeuroSync | Real-Time AI Meeting Intelligence and Autonomous CRM

Stack: OpenAI Whisper, Speaker Diarization, LangGraph, Salesforce API, AWS Lambda, API Gateway

- Designed a real-time meeting intelligence platform that transcribes live meetings via OpenAI Whisper with speaker diarization, extracts action items, deal signals, and sentiment, and syncs to Salesforce CRM within 3 seconds.
- Deployed a fully serverless, event-driven architecture on AWS Lambda and API Gateway sustaining sub-200ms processing latency per transcript chunk at concurrent scale.

Project: SynthAgent | Autonomous AI Research Analyst (Self-Correcting ReAct)

Stack: ReAct, Self-Reflection Loops, Vertex AI Gemini, LangGraph, Firecrawl

- Built a ReAct-pattern research agent with self-correction loops that autonomously plans research, evaluates its own output quality, triggers additional tool calls, and refines findings before delivering citation-rich reports via Vertex AI Gemini.
- Reduced senior-analyst research cycles from 8 hours to under 25 minutes through multi-source Firecrawl and RAG synthesis.

Senior AI Engineer

Apr 2023 - Mar 2025

Gicoh | Lahore, Pakistan | Full-time, On-site

Project: Real-Time Live-Call Conversational AI Chatbot

Stack: RAG, LLaMA 3.2, FAISS, Google STT / TTS, Python SIP, Elastix VoIP PBX

- Architected an end-to-end real-time conversational AI system with a custom RAG pipeline (LLaMA 3.2 and FAISS) delivering sub-second retrieval over live inbound calls, integrating a custom Python SIP client with an Elastix VoIP PBX for autonomous 24/7 call handling.

Project: HighTribe | Multi-Agent Travel Assistant (Live on App Store and Google Play)

Stack: LangGraph, Multi-Agent, Gemini, GPT-4, Google Maps API | Live: hightribe.com

- Designed a production multi-agent LangGraph system live on web, Google Play, and the App Store, with specialized agents for internal database querying, Google Maps, and real-time web search, integrating dual-LLM routing (Gemini and GPT-4) with fallback strategies for quality and cost efficiency.

Project: DocuMind | Autonomous AI Legal Document Review Agent

Stack: Azure OpenAI GPT-4o, LangGraph, RAG, Named Entity Recognition (NER), LegalTech

- Architected an autonomous legal-review agent ingesting contracts and NDAs to flag risky clauses, missing obligations, and jurisdiction-specific issues via LangGraph multi-step reasoning over a legal RAG knowledge base (Azure OpenAI GPT-4o), delivering risk-severity reports that cut law-firm review from days to under 10 minutes per document.

Project: VisionGuard | Multimodal AI Fraud Detection (FinTech)

Stack: Multimodal AI, Anomaly Detection, AWS SageMaker, Computer Vision, Behavioral Biometrics

- Built a fraud-detection system fusing transaction data, document image analysis (ID and cheque verification), and behavioral biometrics into a unified risk score via a multimodal transformer, deployed on AWS SageMaker auto-scaling endpoints handling 10,000+ transactions per minute at p99 latency under 80ms.

Project: CareVision | HIPAA-Compliant Healthcare AI Management System

Stack: Healthcare AI, HIPAA, Voice AI, NLP, PHI Security, RBAC | Live: carevisioncms.co.uk

- Developed a HIPAA-compliant healthcare platform with PHI encryption at rest and in transit, RBAC, and audit logging meeting US federal data-privacy requirements, plus an AI care-plan generation engine using LLMs with voice-based data entry (STT / TTS) and NLP medical entity extraction from clinical notes.

Additional systems delivered at Gicoh

Stack: n8n, GPT-4, LLaMA, LangGraph, Natural Language to SQL, YOLOv8, DeepSORT, FastAPI, RTSP

- IntelliFlow: architected an enterprise workflow automation platform on n8n integrated with LLM-powered agents (GPT-4, LLaMA), using modular agent pipelines with decision-making, conditional branching, and dynamic task execution across enterprise systems.
- CRM / ERP Data Extraction Agent: built a LangGraph agent with a custom Natural Language to SQL tool letting non-technical users query production ERP databases in plain English. Employee Counting and Tracking: built a real-time CCTV computer vision system (FastAPI, RTSP) using YOLOv8 and DeepSORT for persistent identity tracking, plus a face-recognition layer for named-presence detection.

AI Engineer

Dec 2020 - Apr 2023

Techling (Private) Limited | Lahore, Pakistan | Full-time, On-site

Project: Chex.AI | End-to-End Vehicle Inspection AI Platform

Stack: YOLOv8, AWS, FastAPI, Pixtral Multimodal LLM, Amazon S3 | Live: chex.ai

- Delivered a full-stack AI vehicle-inspection platform (FastAPI, AWS authentication, S3) with license plate extraction and vehicle-damage detection using fine-tuned YOLOv8, integrating the Pixtral multimodal LLM for human-readable damage-assessment reports plus a video object-tracking module for live streams.

Project: DeftGPT | Multi-LLM Unified AI Platform (11 LLMs)

Stack: FastAPI, AWS, Multi-LLM Orchestration, GPT-4, LLaMA | Live: deftgpt.com

- Built and deployed a unified platform integrating 11 LLMs through a single FastAPI backend on AWS, architecting LLM API routing, authentication, and session management so users could compare, switch, and receive parallel responses across models.

Trainee Data Scientist

Aug 2020 - Nov 2020

Programmers Force | Lahore, Pakistan | Full-time, On-site

SELECTED INDEPENDENT PROJECTS, R&D AND BENCHMARKING

- Multi-Mode AI SaaS Workspace (MVP architecture): designed a FastAPI and LangGraph workspace with four specialized assistant modes and PostgreSQL-backed conversation checkpointing, covering data model, streaming API surface, and phased delivery plan.
- Multi-Tenant WhatsApp Business Cloud API Chatbot: architected a multi-tenant messaging platform including template-message pricing models, Tech Provider onboarding, and per-tenant session routing.
- Self-Hosted Open-Weight LLM Feasibility Study: benchmarked open-weight candidates (Qwen3, GLM-4.5-Air, gpt-oss-120b) against a production agent workload, sizing GPU VRAM for FP8 and Mixture-of-Experts variants, comparing vLLM serving behind an OpenAI-compatible endpoint, and assessing data residency.
- LLM Cost and Token Economics Benchmarking: profiled a 37-tool agentic ReAct loop and established it as 99.7% input-bound with the tool schema (~38K tokens) as the dominant cost driver, then quantified savings from prompt caching and tool-schema pruning across competing model families.
- Document AI and RAG Ingestion Strategy: evaluated PDF parsing stacks including Docling for table-heavy technical documents, comparing layout fidelity, chunking behavior, and retrieval accuracy.

EDUCATION

MS Data Science (Artificial Intelligence / Machine Learning)

Feb 2020 - Jan 2023

AUIC Islamabad, Pakistan | CGPA 3.12 / 4.0

BS Computer Science

Sep 2015 - Aug 2019

GC University Faisalabad, Pakistan | CGPA 3.42 / 4.0

CERTIFICATIONS

Machine Learning Specialization | Stanford University and DeepLearning.AI (Coursera), Andrew Ng